

Major Project Presentation

An Interactive Real-Time System for Tracking and Analyzing Weather and Pollution Patterns

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Objective

Most existing weather and air quality applications provide only real-time data — they lack **historical analysis, forecasting, and collaborative features** within a single platform, forcing users to depend on multiple separate applications for complete environmental information.

The objective of **ClimaSphere** is to provide a **unified platform for real-time weather monitoring, air quality analysis, and environmental forecasting** — addressing these limitations by integrating **real-time data, historical analysis, predictive insights, knowledge sharing, and collaborative features** into a single application. It is designed to assist the **general public, students, scientist, researchers, environmental analysts, educators, NGOs, government agencies, and environmental activists** in understanding environmental conditions, promoting awareness, supporting research, and making **informed, data-driven decisions**.



ABSTRACT

ClimaSphere is a full-stack web-based environmental intelligence platform developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)**. The system integrates the **OpenWeather Weather API** and **Air Pollution API** to provide real-time monitoring of **temperature, humidity, wind speed, AQI, PM2.5, PM10, NO₂, SO₂, and CO** for locations worldwide. Historical datasets from 20+ Indian cities are analysed using Python for Exploratory Data Analysis and trend identification. Prophet-based forecasting predicts future AQI and weather conditions, deployed via Streamlit. Authenticated users can publish analytical insights, download CSV datasets, and collaborate through community chat, voice calls, and social blogging, making ClimaSphere a unified platform for environmental monitoring, analysis, and awareness.

COMPARATIVE TABLE

Feature	Google Weather	IQAir	OpenWeather	Govt Portals	ClimaSphere
Real-time Weather Data	✓ Yes	✗ Limited	✓ Yes	✗ No	✓ Yes
Real-time AQI Data	✗ Limited	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Weather + AQI Single Platform	✗ No	✗ Limited	✓ Yes	✗ No	✓ Yes
AQI Based on Multiple Pollutants	✗ No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Historical Data Access	✗ No	✗ Limited	⚠ API Only	✓ Yes	✓ Yes
CSV Data Download	✗ No	✗ No	⚠ API Only	✗ Limited	✓ Yes
Exploratory Data Analysis (EDA)	✗ No	✗ No	✗ No	✗ No	✓ Yes
AQI Forecasting	✗ No	✗ Limited	✗ No	✗ No	✓ Yes
Weather Forecasting	✓ Yes	✗ Limited	✓ Yes	✗ Limited	✓ Yes
Analysis & Insights Repository	✗ No	✗ No	✗ No	✗ No	✓ Yes
Community Chat & Collaboration	✗ No	✗ No	✗ No	✗ No	✓ Yes
User Authentication & Profiles	✗ Limited	✓ Yes	⚠ API Only	✗ No	✓ Yes
Student / Research Platform	✗ No	✗ No	✗ No	✗ No	✓ Yes

LITERATURE REVIEW

Environmental monitoring has become a critical research area due to rising concerns about **climate change, urbanization, and air pollution**. Recent studies by **Grech et al. (2024)** and **Chadalavada et al. (2025)** highlight that while modern platforms provide **real-time data** and **AI-based forecasting**, most suffer from fragmented features — focusing either on monitoring or prediction, rarely combining both within a single system. Existing platforms such as **IQair, OpenWeather, Windy, and AccuWeather** primarily serve specific functionalities like weather display or AQI reporting. They lack **analytical & insights repository, historical data exploration, research collaboration, and knowledge-sharing capabilities** — forcing users to switch between multiple platforms for complete environmental information. **ClimaSphere** addresses this gap by combining **real-time monitoring, Exploratory Data Analysis, Prophet-based forecasting, an Analysis and Insights Repository, and community collaboration** into a single unified web-based platform.

USED TECH STACK

Frontend: React.js v19, Tailwind CSS v4, Chart.js, Vite v7

Backend: Node.js v22, Express.js v5

Database: MongoDB v8, Mongoose v8

Real-Time Communication: Socket.IO v4 (*Chat, Notifications*)

Authentication & Security: JWT (JSON Web Token), bcrypt

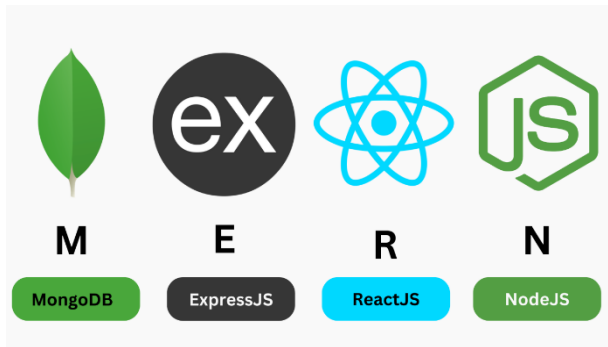
Media Storage: Cloudinary SDK v2, Multer v2

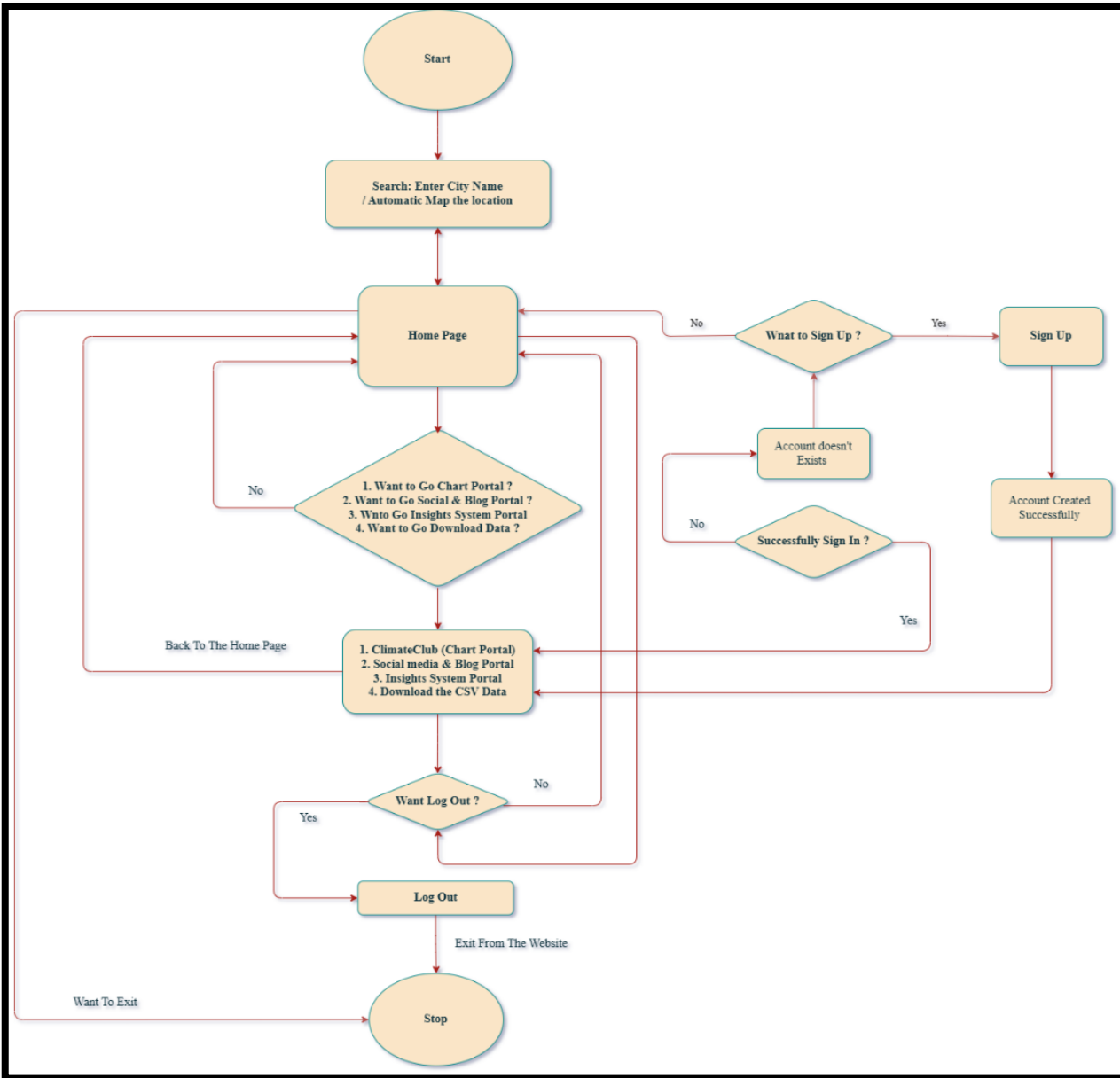
Python & ML Libraries: Pandas, NumPy, Matplotlib, Seaborn, Poltly, Scikit-learn, Prophet Model, Joblib

Forecasting Deployment: Streamlit (*AQI & Weather Forecasting Apps*)

Frontend Deployment: Vercel

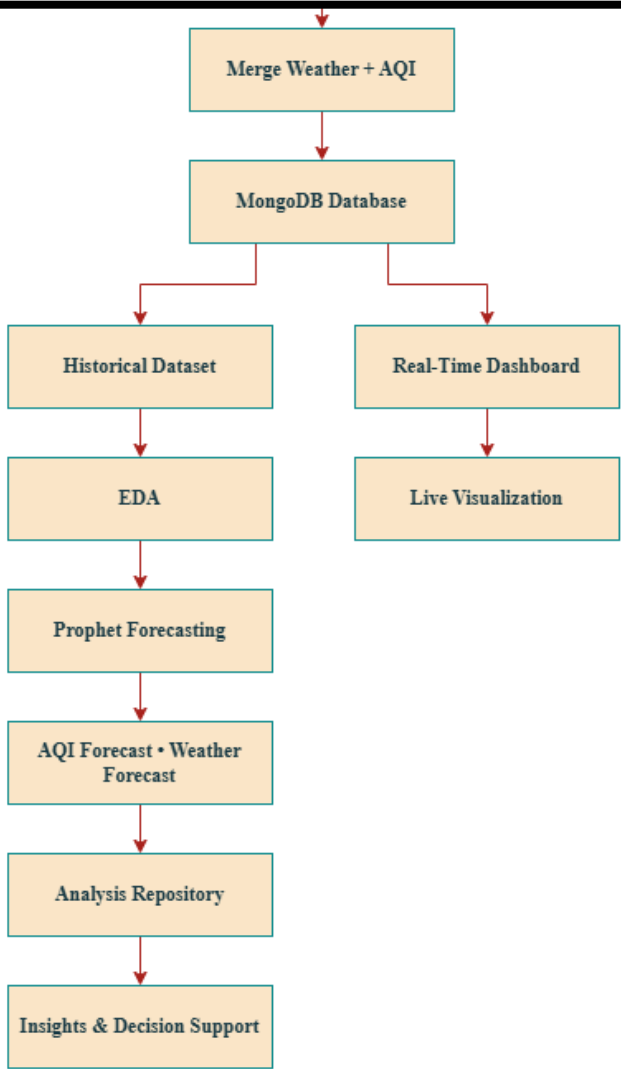
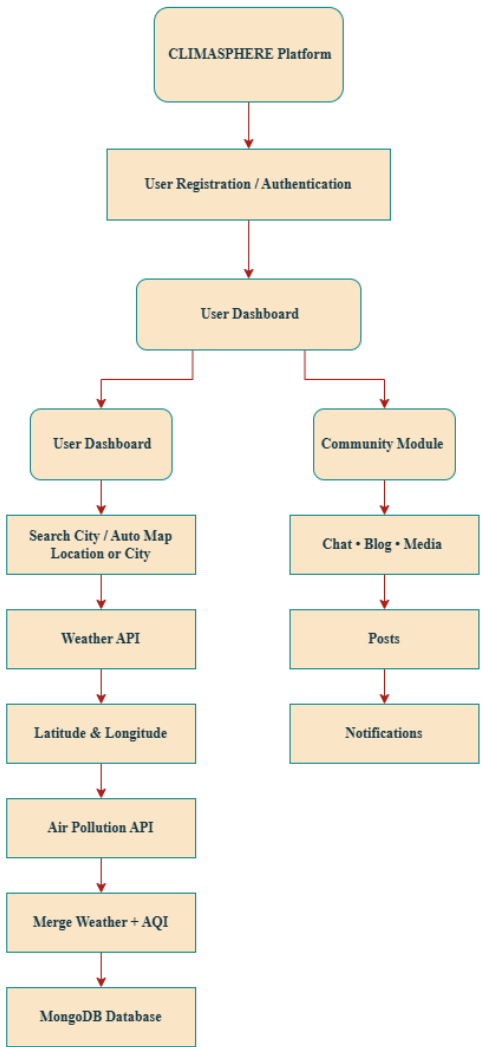
Development Tools: Postman, Git, GitHub, VS Code, Jupyter Notebook





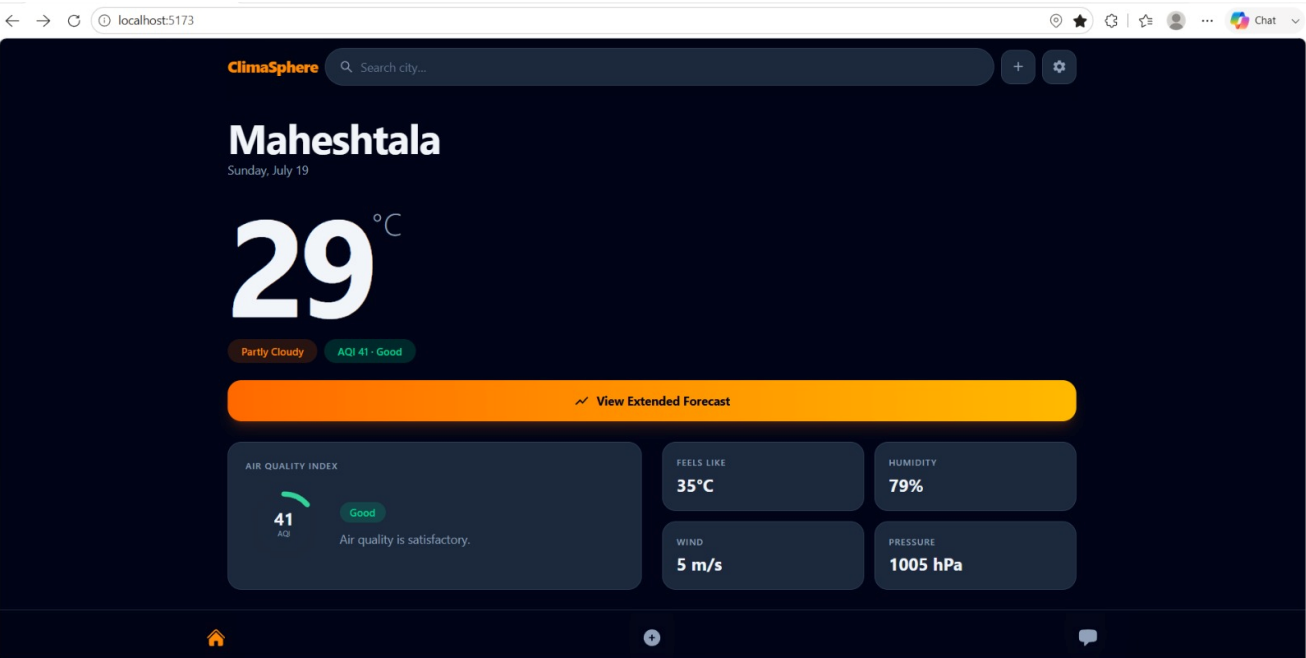
- **Step 1 — User Authentication:** JWT-secured login, registration, and role-based access control (*User / Analyst / Admin*)
- **Step 2 — Environmental Data Acquisition:** City search or auto-location detection → **OpenWeather Weather API** fetches live data → coordinates used to call **Air Pollution API**
- **Step 3 — Data Processing & Storage:** Weather and AQI responses merged, validated, and stored in **MongoDB**
- **Step 4 — Real-Time Dashboard:** Processed data displayed via **React.js** with interactive charts using **Chart.js**
- **Step 5 — Historical Data & EDA:** Historical Indian city datasets analysed using **Python** (*Pandas, Seaborn, Matplotlib*)
- **Step 6 — Forecasting: Prophet model** generates AQI and weather forecasts, deployed via **Streamlit**.
- **Step 7 — Insights & Collaboration:** Analysts publish findings via **Analysis & Insights Repository**; users collaborate through **chat**, and **blogging**
- **Step 8 — Alerts & Notifications:** Real-time weather alerts and AQI notifications delivered to users via **Socket.IO**

Flow Chart of System Development



OVERVIEW OF USER INTERFACE

Home Page



Sign in & Sign up page

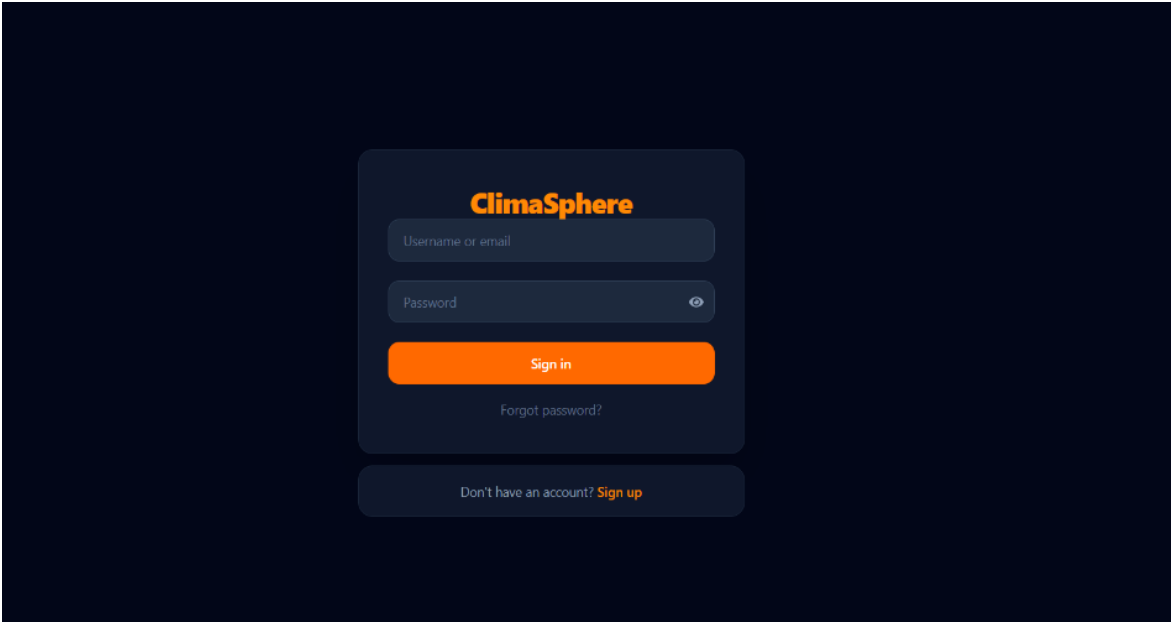
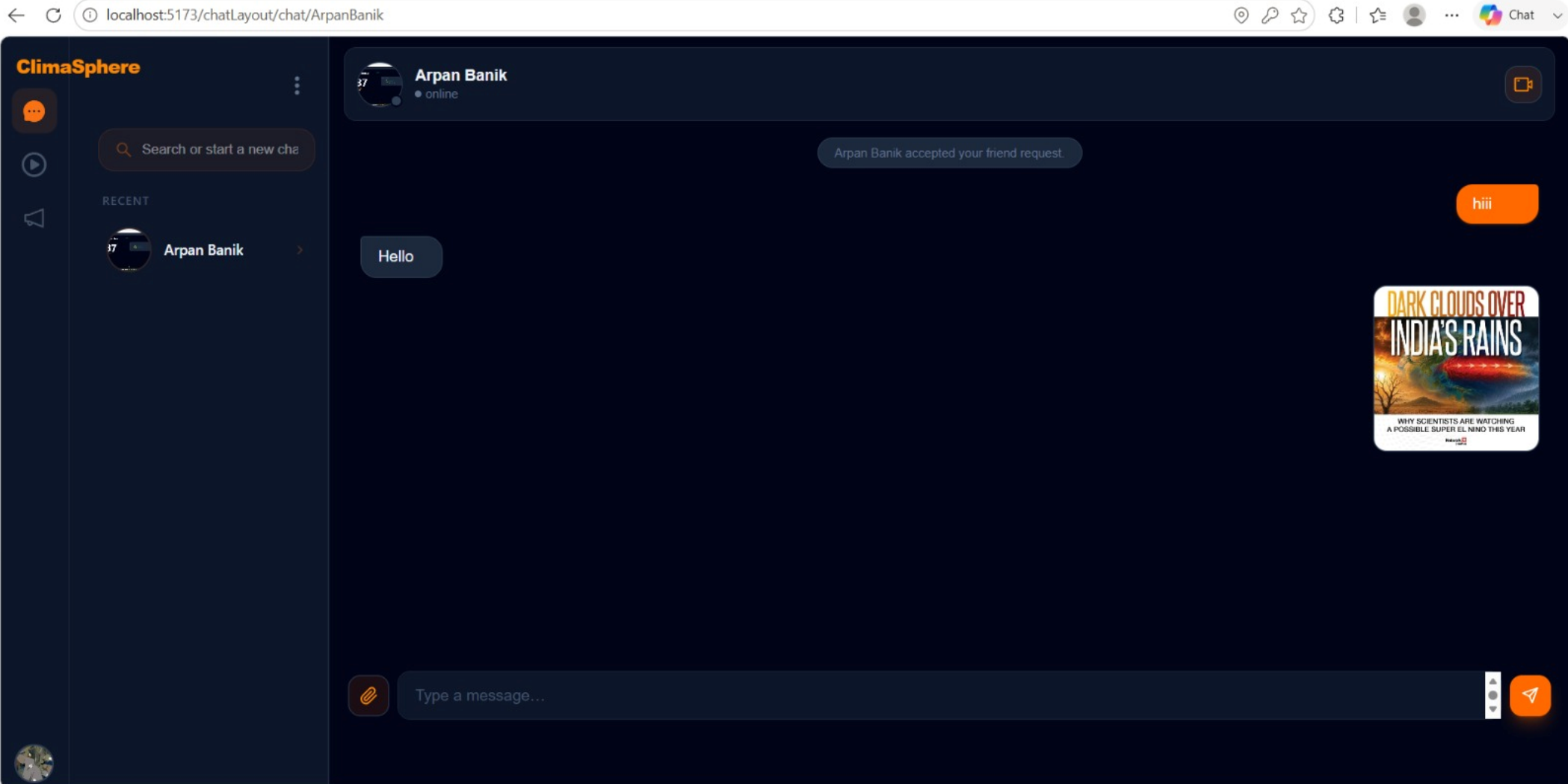
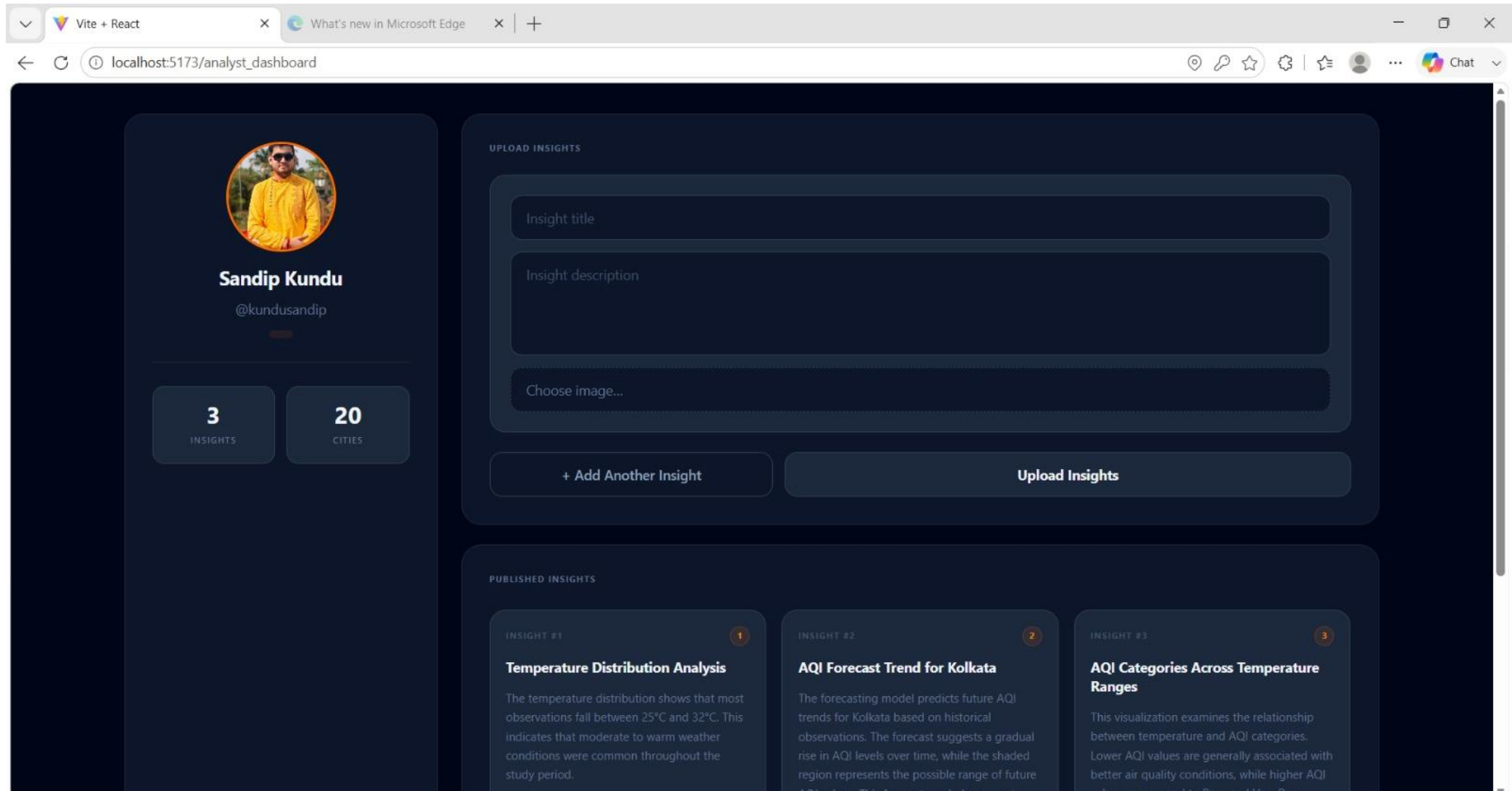


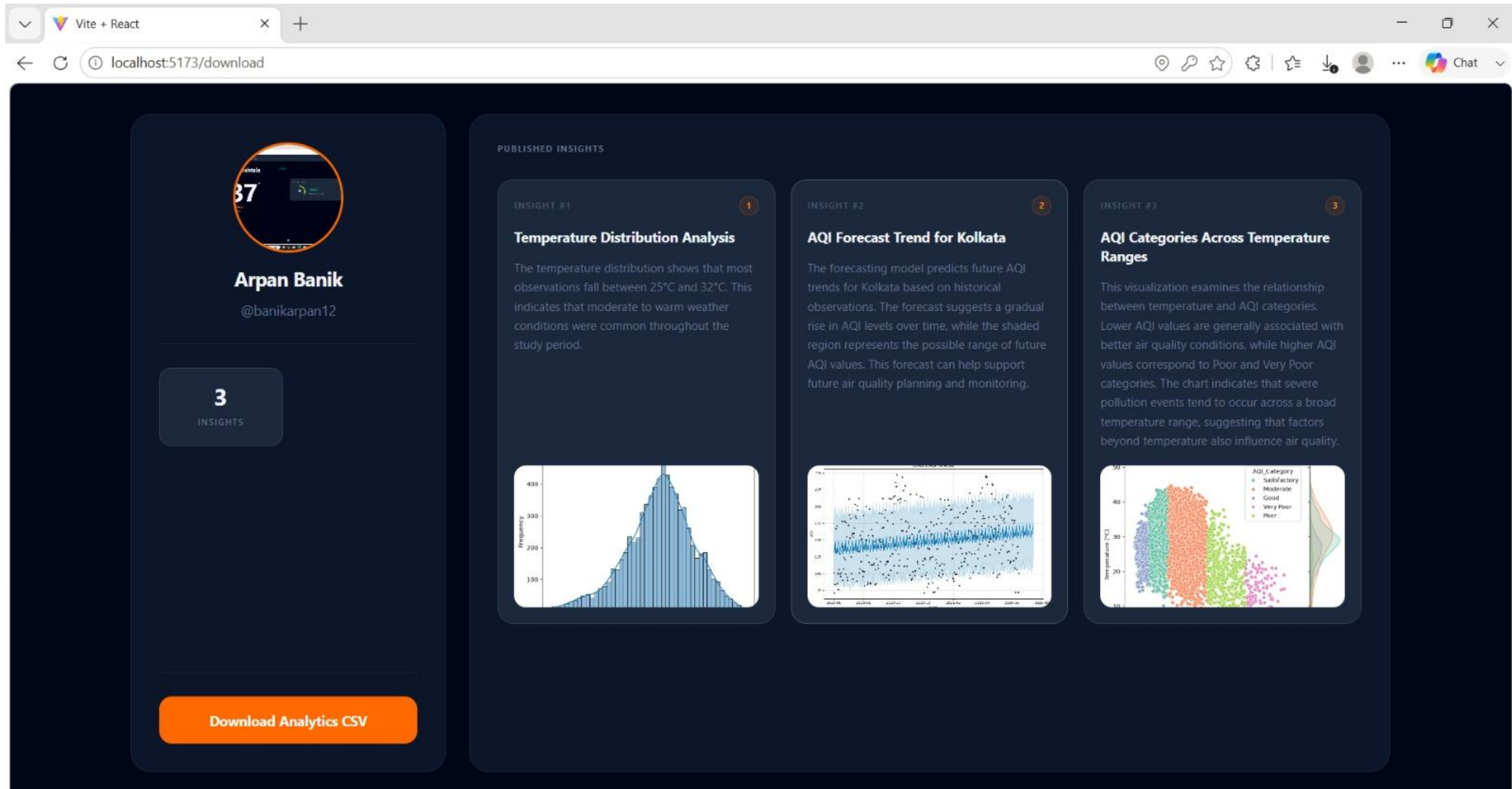
CHART SYSTEM



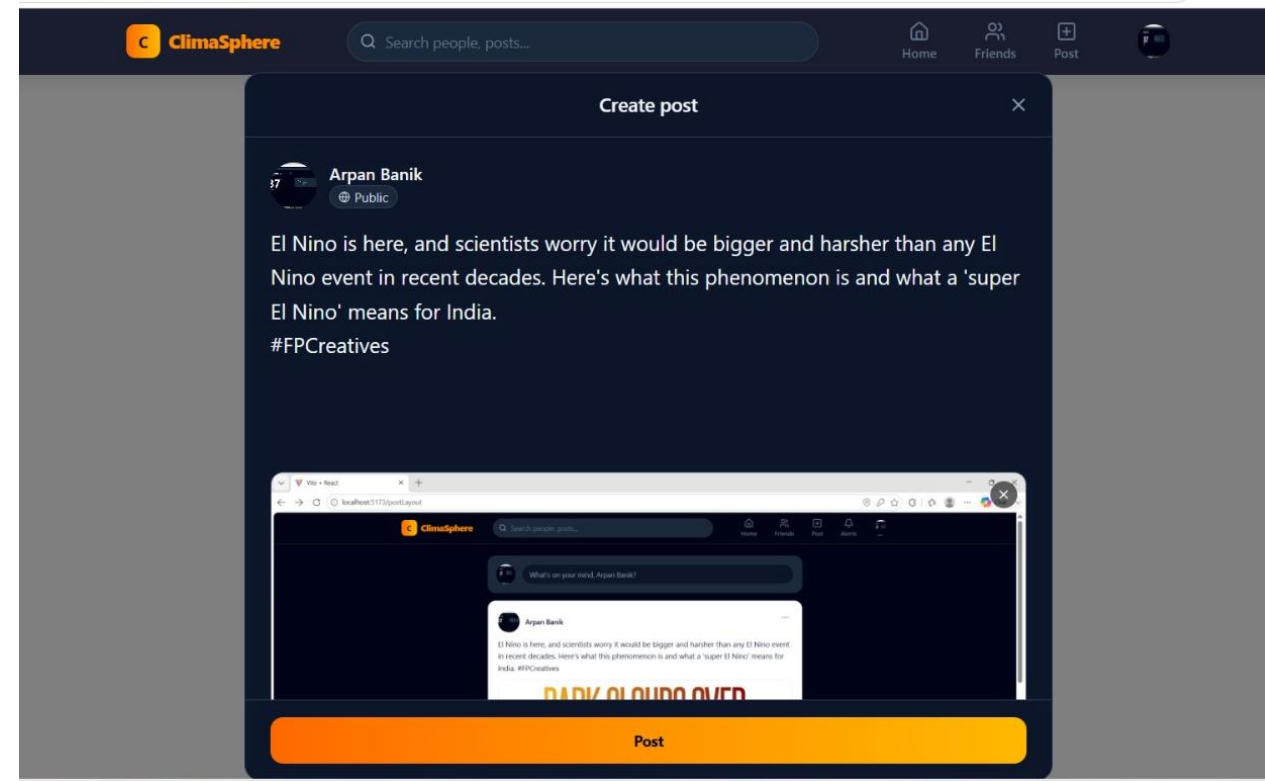
ANALYST REPOAITORY



ANALYSIS & INSIGHTS REPOAITORY



SOCIAL MEDIA & BLOG MODULE



AQI FORECASTING SYSTEM

INDIA FORECASTING HUB

All Systems

AQI Forecasting

Weather Forecasting

INDIA AIR QUALITY INTELLIGENCE

 AQI Forecasting System

Prophet-powered time-series forecasting · 20 Indian cities · Up to 365-day predictions

CONFIGURE YOUR FORECAST

Select City

Ahmedabad

Select Forecast Date

2026/07/19

Forecast AQI

Today: 19 Jul 2026 · Latest available data for Ahmedabad: 14 Jul 2026 · forecasting 5 days ahead to today

Ahmedabad Historical Overview

TEMPERATURE

29.7°C

Historical Average

HUMIDITY

41.9%

Historical Average

PM2.5

62.8

µg/m³

AVERAGE AQI

125

Moderate

WEATHER

Clear

Most Frequent

RECORDS

560

560 Days

WEATHER FORECASTING SYSTEM

INDIA FORECASTING HUB

All SystemsAQI ForecastingWeather Forecasting

INDIA WEATHER INTELLIGENCE

Weather Forecasting System

Prophet-powered time-series forecasting · 20 Indian cities · Temperature, Humidity, Pressure & Wind

Select CityAhmedabadJump to dayToday

Model trained through 14 Jul - forecast projected forward to today

Ahmedabad

Sunday, 19 July 2026

19° Clear

Feels like 17°C

Clear skies throughout. Feels like 17°C, wind 5 km/h, humidity 46%.

HUMIDITY46%

WIND4.6 km/h

GUST9.6 km/h

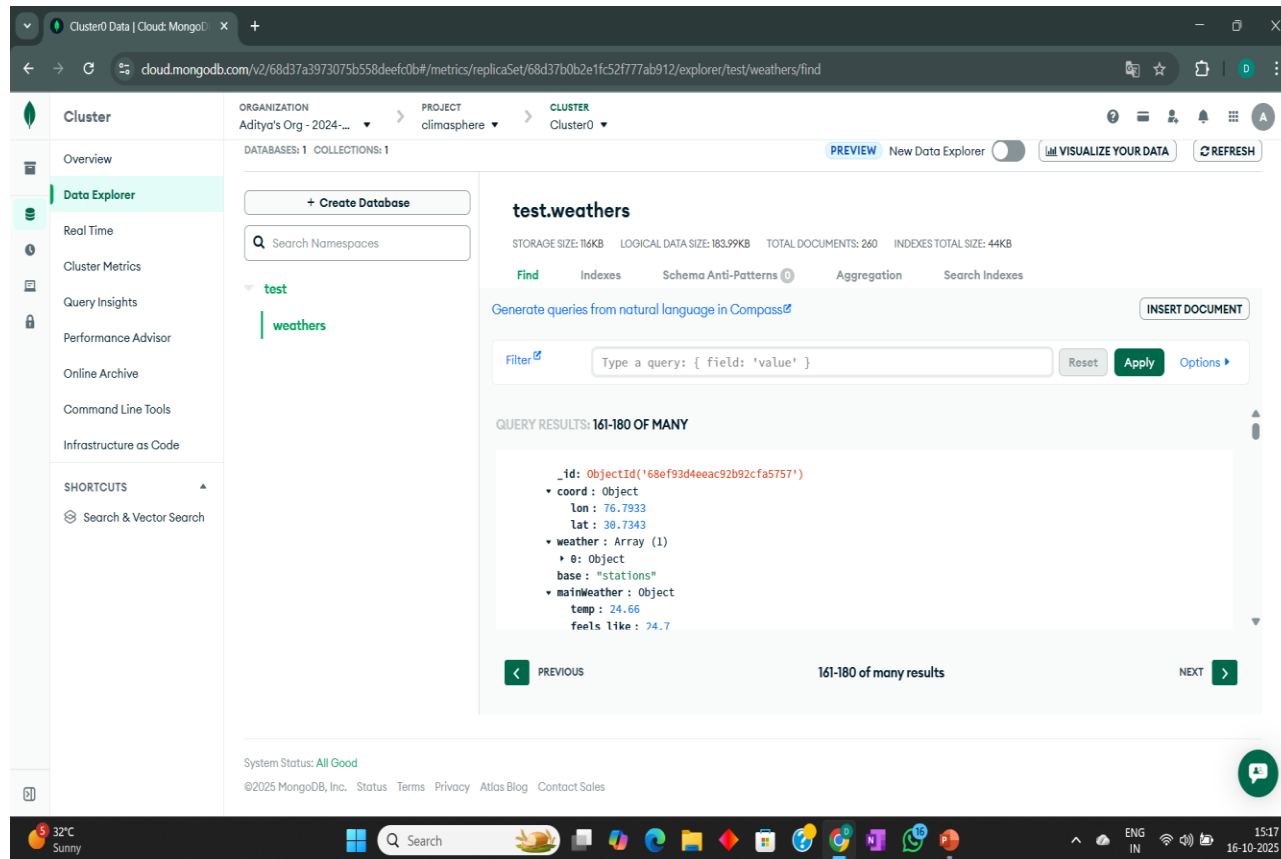
PRESSURE1011 hPa

RAIN4%

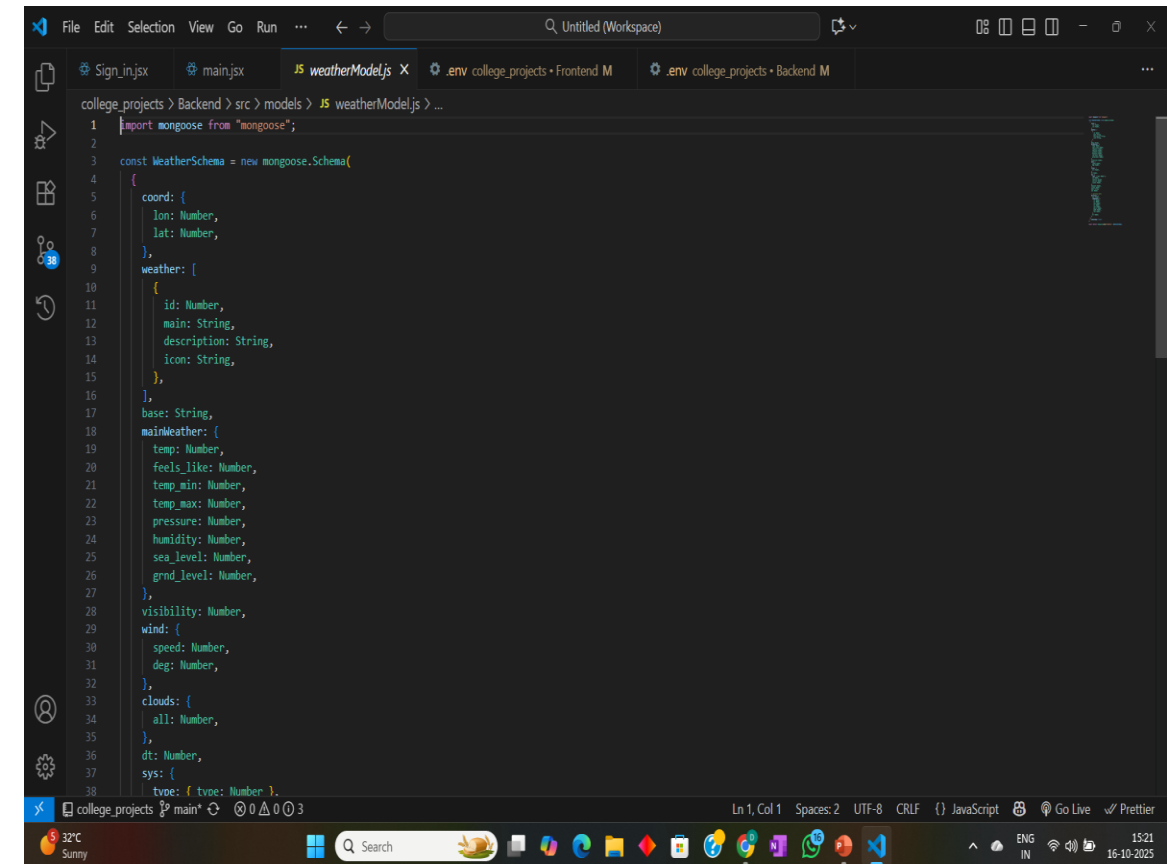
UV INDEX0

OVERVIEW OF BACKEND SURFACE

Location wise database of weather and pollution data



Mongoose schema of weather and pollution data



EXPLORATORY DATA ANALYSIS

Weather Analysis: Temperature Distribution

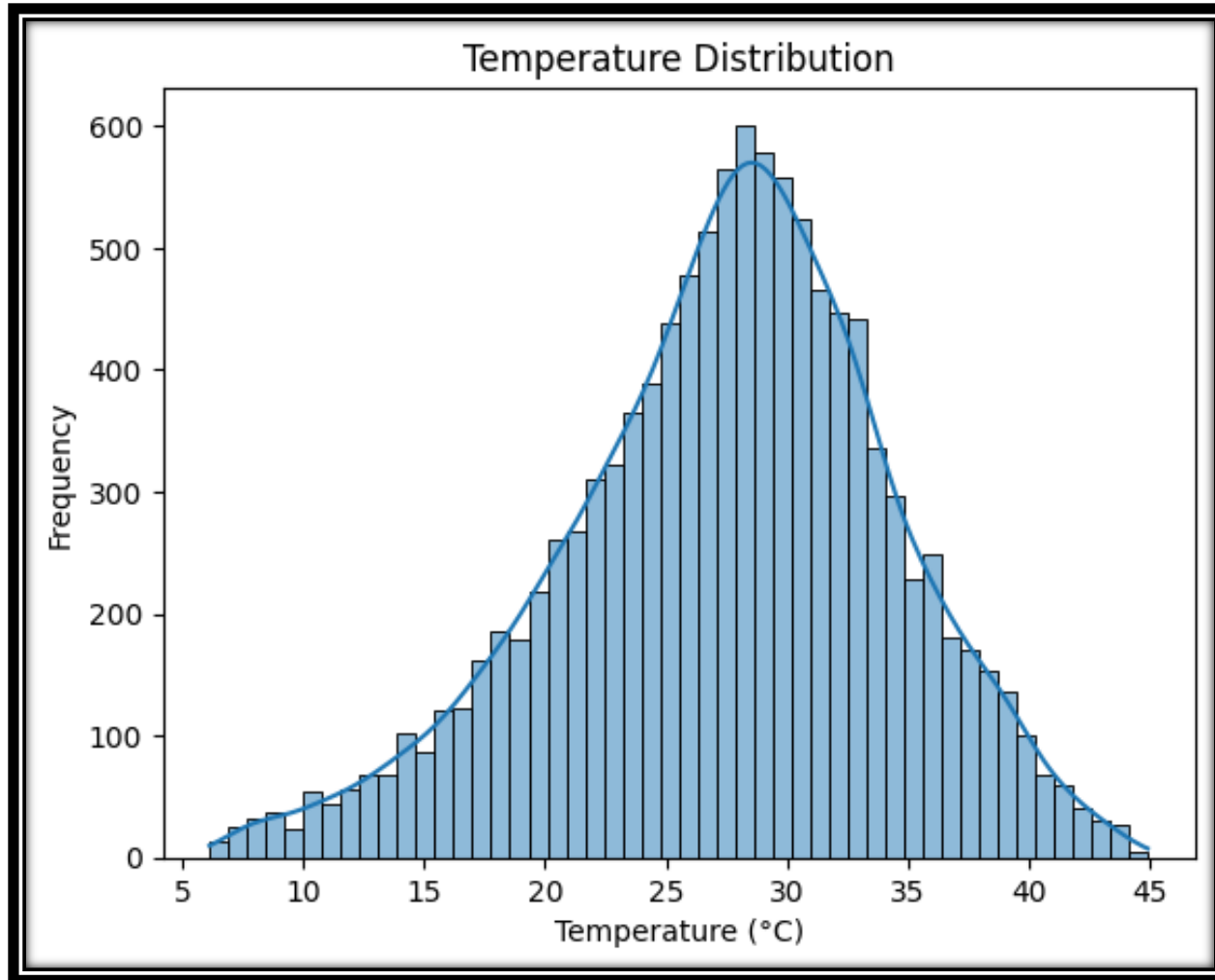


Chart Type: Histogram with Kernel Density Estimation (KDE)

Description: The chart shows the distribution of temperature values (°C) across the dataset, indicating how frequently different temperature ranges occur.

Key Insights:

- Most temperature values are concentrated between 24°C and 34°C.
- The highest frequency occurs around 29°C–30°C, representing the most common temperature range.
- Very low (<10°C) and very high (>40°C) temperatures occur rarely.
- The smooth bell-shaped distribution indicates a consistent temperature pattern.

Weather Analysis: Top 5 Hottest & Coldest Cities

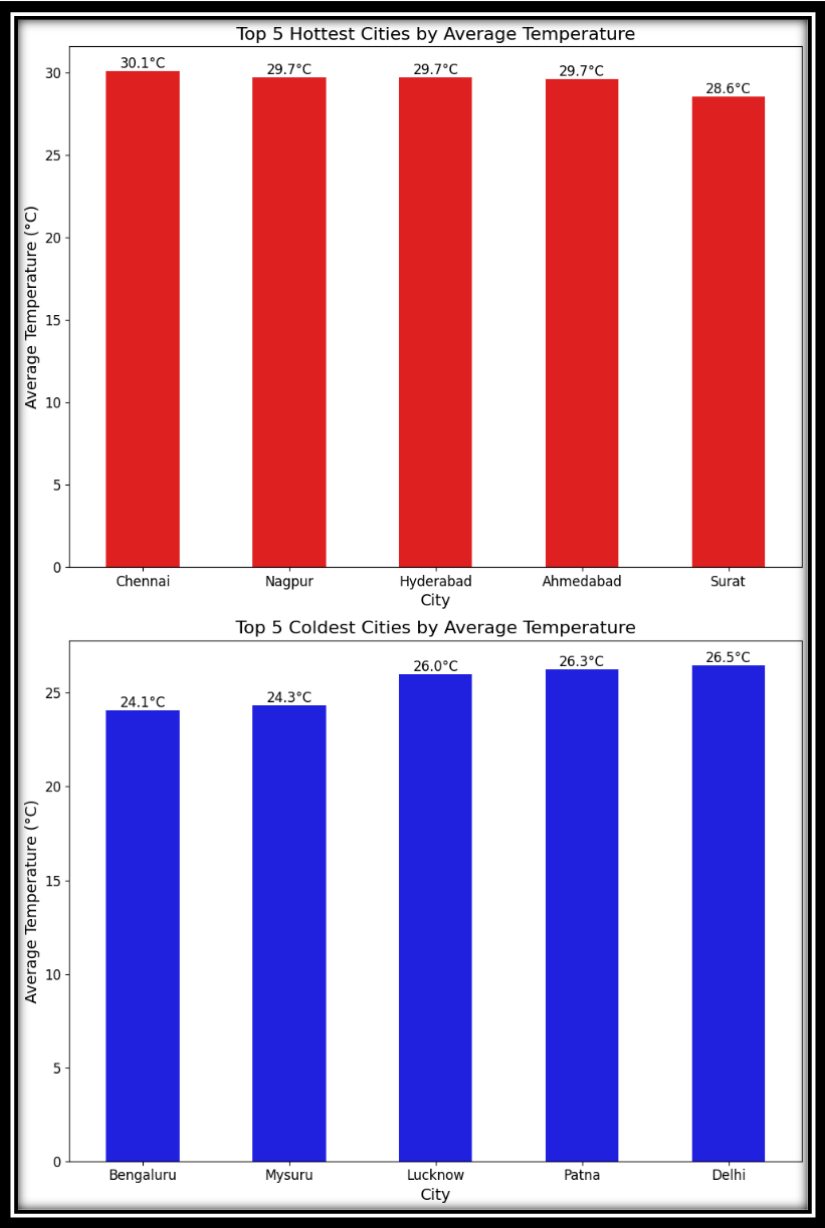


Chart: Top 5 Hottest & Coldest Cities by Average Temperature

Key Insights:

The bar charts compare the **hottest and coldest cities** by average temperature across the dataset. **Chennai** recorded the highest average temperature at **30.1°C**, followed closely by **Nagpur, Hyderabad, and Ahmedabad** at **29.7°C** each, and **Surat** at **28.6°C** — with relatively small differences among the top five. On the other end, **Bengaluru** recorded the lowest average temperature at **24.1°C**, followed by **Mysuru (24.3°C)** and **Lucknow, Patna, and Delhi** ranging between **26.0°C and 26.5°C**. Overall, the temperature variation among both the hottest and coldest cities is relatively small, reflecting **consistently warm climatic conditions** across Indian cities with only moderate regional differences.

EXPLORATORY DATA ANALYSIS

Pollution Analysis: Temperature Distribution

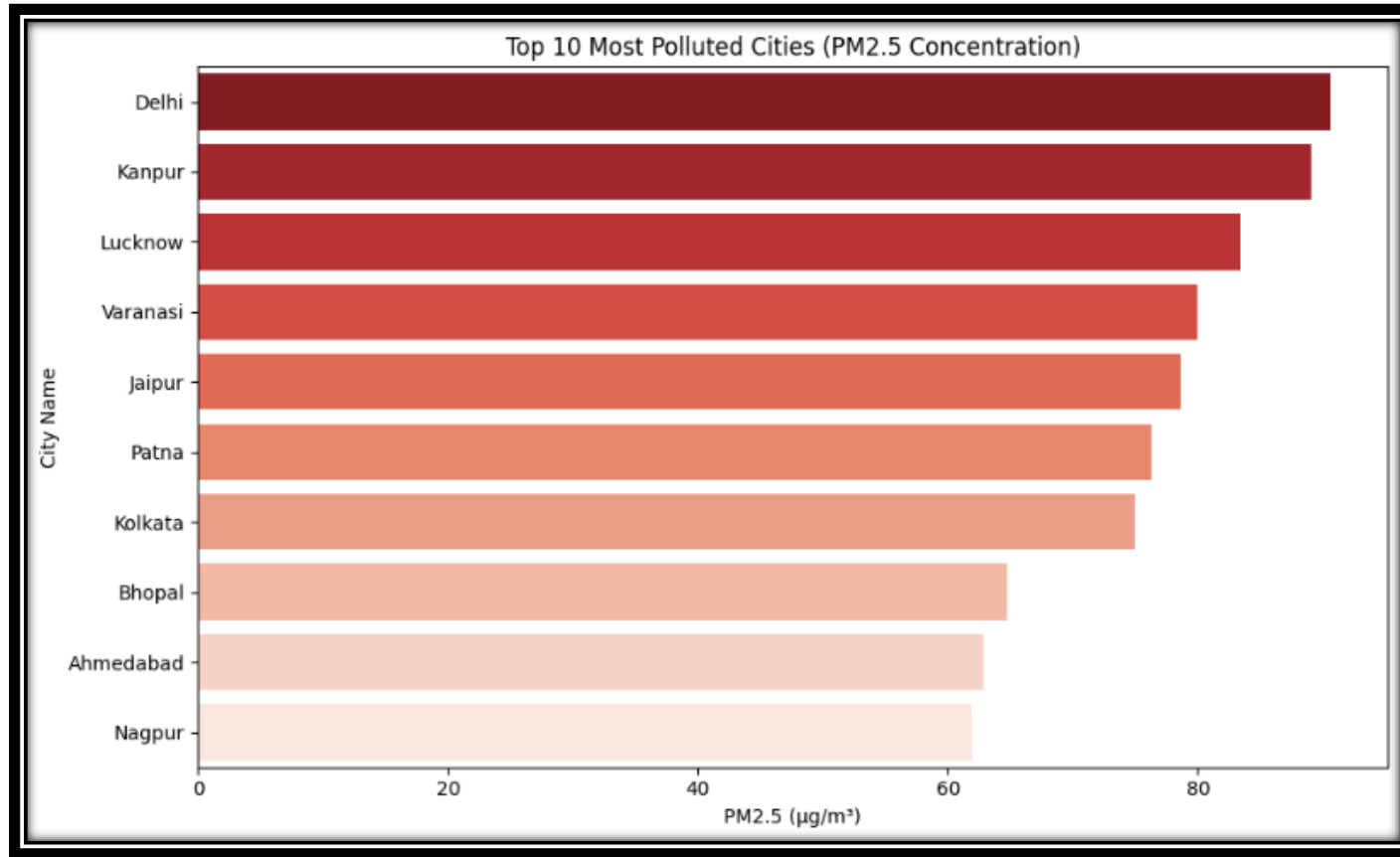


Chart Type: Horizontal bar chart

Description: The chart displays the **top 10 most polluted cities** based on their average **PM2.5 concentration ($\mu\text{g}/\text{m}^3$)**, highlighting cities with the highest particulate matter levels.

Key Insights:

- Delhi recorded the highest PM2.5 concentration, making it the most polluted city in the dataset.
- Kanpur and Lucknow ranked second and third, respectively, with consistently high PM2.5 levels.
- Varanasi, Jaipur, Patna, and Kolkata also recorded elevated PM2.5 concentrations.
- Nagpur recorded the lowest PM2.5 concentration among the top 10 most polluted cities

Pollution Analysis: Top 5 Cities by Avg. PM2.5 & PM10

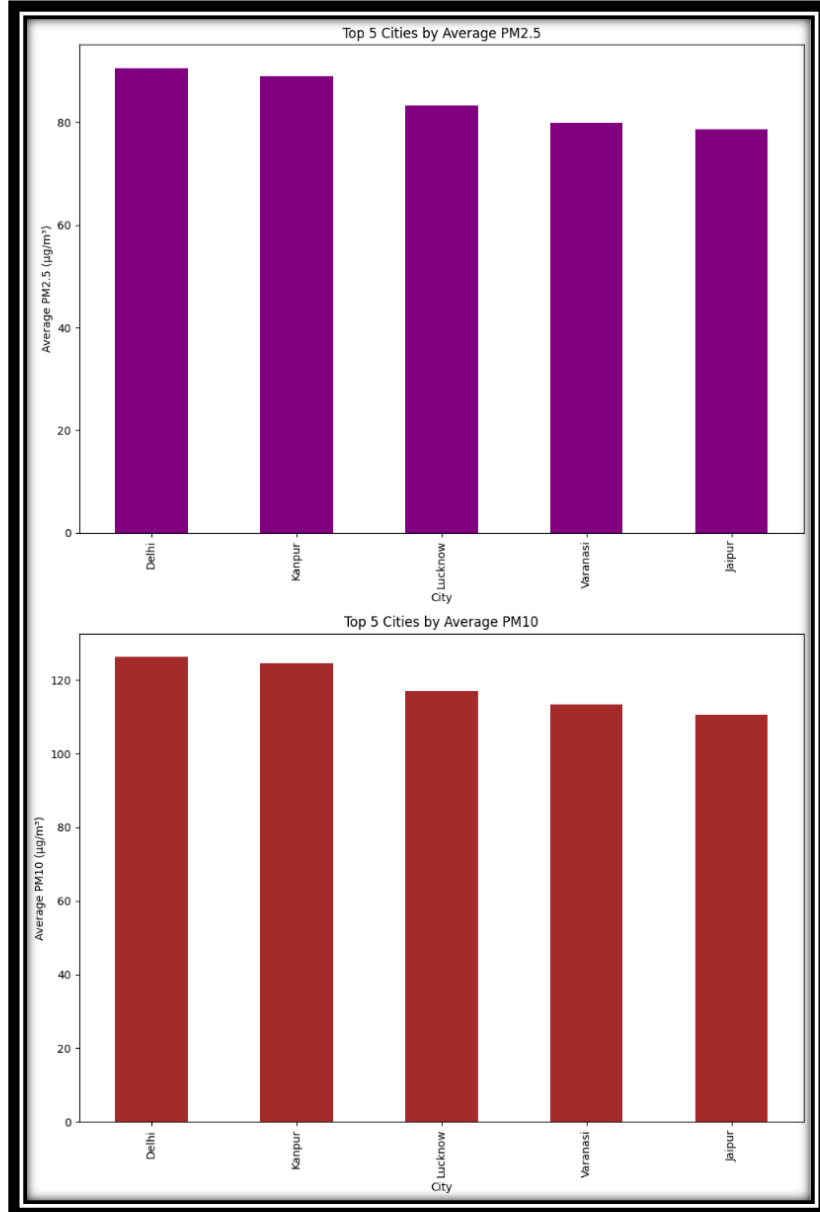


Chart: Top 5 Cities by Average PM2.5 & PM10

Key Insights:

The charts illustrate the **PM2.5 concentration levels** across Indian cities and compare the **top cities by average PM2.5 and PM10** particulate matter concentrations recorded in the dataset. **Delhi** recorded the highest PM2.5 concentration making it the most polluted city, followed by **Kanpur** and **Lucknow** in second and third positions respectively. **Varanasi, Jaipur, Patna, and Kolkata** also recorded elevated PM2.5 levels while **Nagpur** recorded the lowest among the top 10. The same ranking pattern is observed for **PM10 concentrations** — confirming a very strong positive relationship between PM2.5 and PM10 (**correlation: 0.99**). The temperature difference among the top polluted cities is relatively small, highlighting that **northern Indian cities** consistently experience the highest particulate matter pollution levels.

EXPLORATORY DATA ANALYSIS

AQI vs. Pollution Analysis: Pollutant Correlation Heatmap

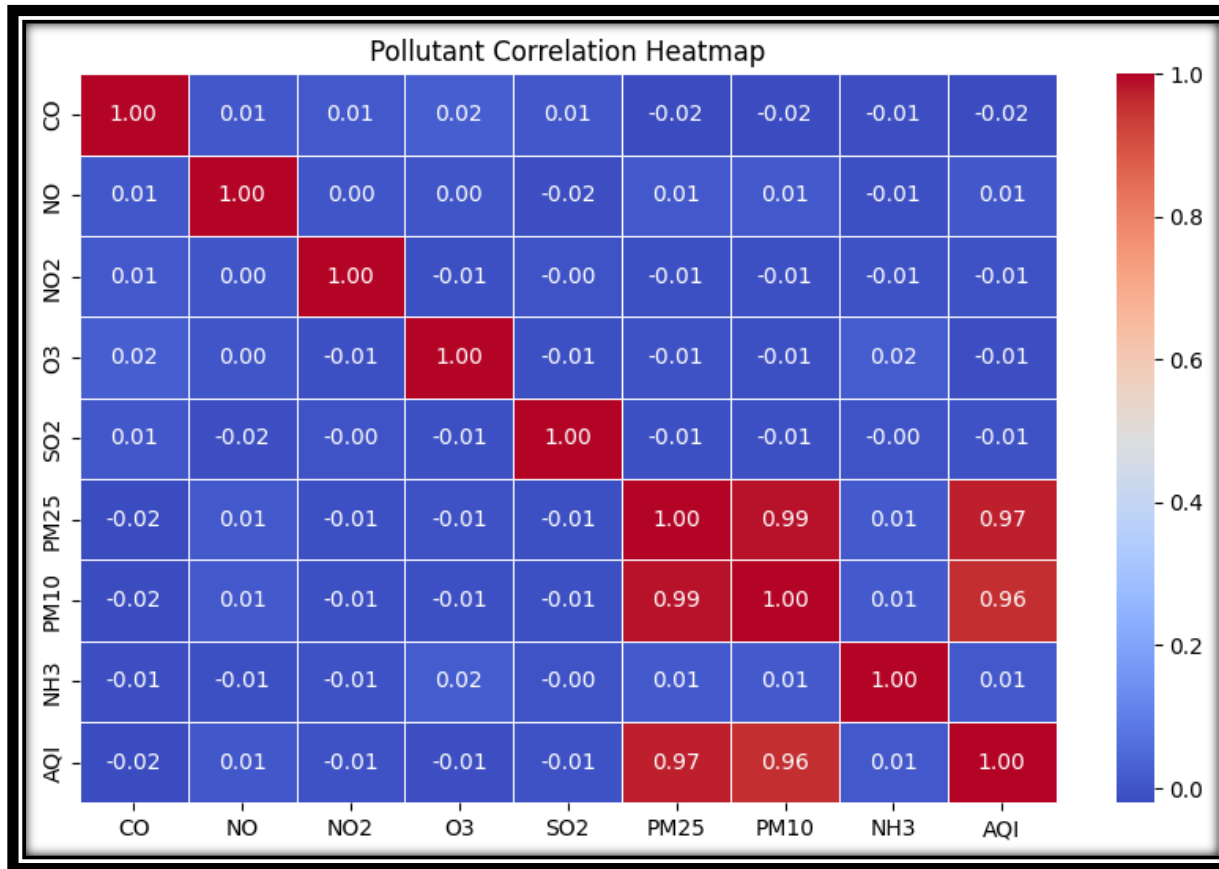


Chart Type: Correlation heatmap

Description: The heatmap shows the correlation between major air pollutants and the AQI, indicating the strength of their linear relationships.

Key Insights:

- PM2.5 and PM10 show a very strong positive correlation (0.99).
- AQI has a strong positive correlation with PM2.5 (0.97) and PM10 (0.96).
- CO, NO, NO₂, O₃, SO₂, and NH₃ exhibit negligible correlations with AQI and with each other.
- Overall, the heatmap indicates that **PM2.5 and PM10 are the pollutants most strongly associated with AQI.**

AQI vs. City Analysis: Top 10 Cities by Avg. AQI

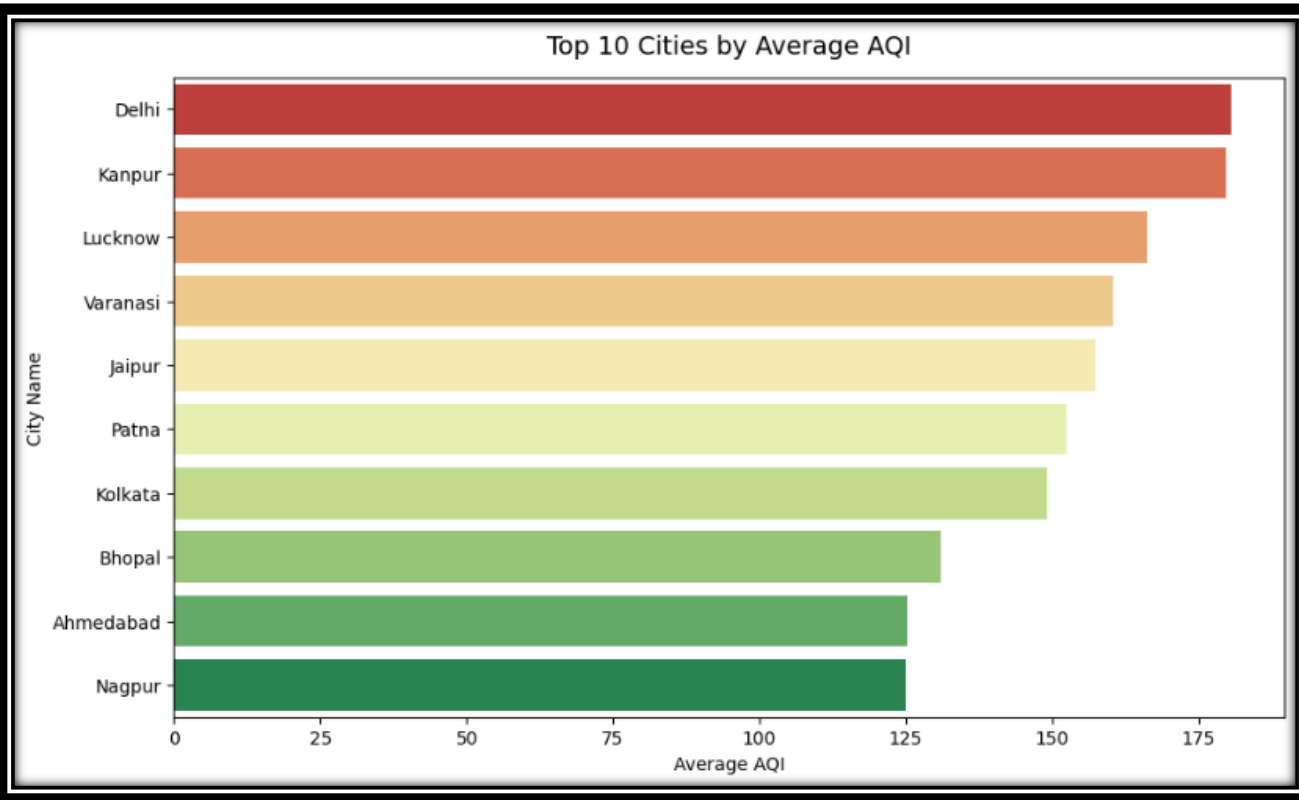


Chart Type: Horizontal Bar Chart

Description: The chart displays the **top 10 cities with the highest average AQI**, highlighting the cities experiencing the poorest overall air quality.

Key Insights:

- Delhi recorded the highest average AQI among all cities.
- Kanpur ranked second, followed by Lucknow and Varanasi.
- Jaipur, Patna, and Kolkata also recorded consistently high average AQI values.
- Nagpur recorded the lowest average AQI among the top 10 cities.
- Overall, the chart indicates that **Delhi and Kanpur** experience the highest average AQI levels.

OVERVIEW OF MODEL OUTCOME

AQI Forecast : AQI Forecast For Kolkata

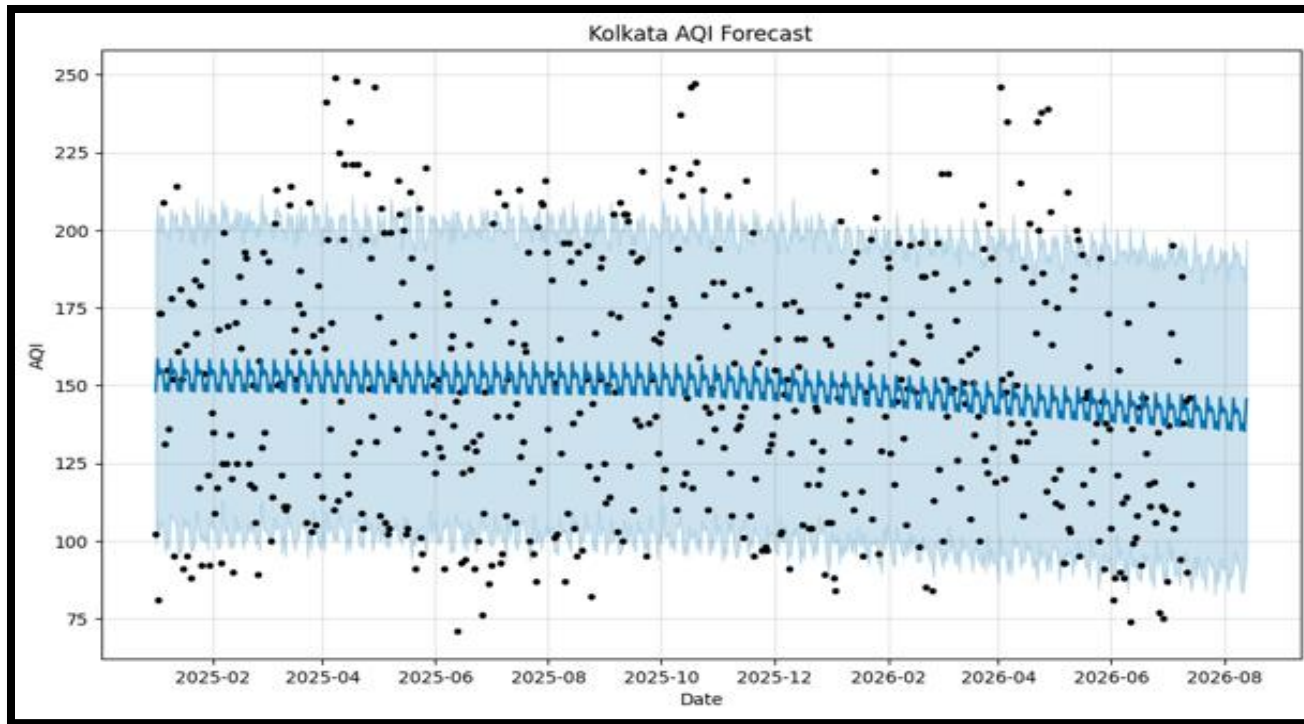


Chart Type: Time Series Forecast Plot (Prophet Model)

Description: The chart presents the **forecasted AQI trend for Kolkata**. The shaded region represents the forecast confidence interval.

Key Insights:

- The forecast indicates a **gradual decline** in AQI levels over the prediction period.
- Most observed AQI values lie within the forecast confidence interval, indicating a reasonable model fit.
- The wider confidence interval in the forecast period reflects **greater uncertainty in long-term AQI predictions**.
- Despite the declining trend, the forecast suggests that AQI levels are expected to remain within the moderate air quality range

OVERVIEW OF MODEL OUTCOME

Weather Forecast : Temperature Forecast For Goa

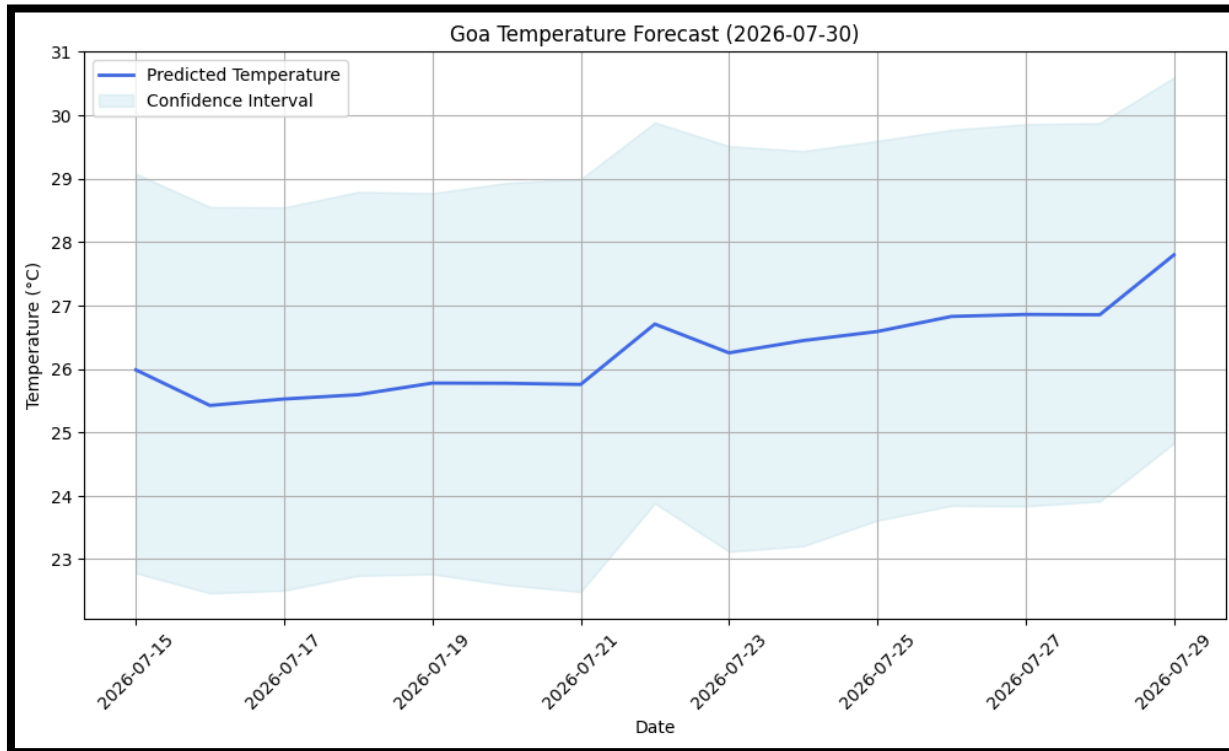


Chart Type: Time Series Forecast Plot (Prophet Model)

Description: The chart presents the **forecasted temperature trend for Goa**. The blue line represents the predicted temperature, while the shaded region indicates the **95% confidence interval** of the forecast.

Key Insights:

- The forecast indicates a **slight upward trend** in temperature during the forecast period.
- Predicted temperatures increase gradually from approximately **26°C to 28°C by 29 July 2026**.
- The forecast confidence interval remains relatively consistent.
- No abrupt fluctuations or extreme temperature changes are predicted during the forecast period.
- The forecast suggests that **Goa is expected to maintain warm and stable weather conditions**, consistent with its coastal climate.

FUTURE SCOPE

Key areas of future work include:

- **Improved Forecasting:** Use advanced machine learning and deep learning models to improve the accuracy of weather and air quality predictions.
- **Scalable Data Storage:** Enhance the system to collect and store environmental data at shorter time intervals using cloud-based infrastructure, enabling long-term analysis and better scalability.
- **Smart Alerts:** Introduce real-time notifications for severe weather conditions and high pollution levels.
- **Enhanced Dashboard:** Add more interactive filters and comparative visualizations for deeper environmental analysis.
- **Wider Accessibility:** Develop a mobile application and support additional languages to make the platform accessible to a larger audience

CONCLUSION

ClimaSphere has been successfully developed as an interactive web-based platform for **real-time weather and air pollution monitoring, environmental data analysis, and forecasting**. By integrating modern web technologies, reliable APIs, data visualization, and machine learning, the system provides users with a comprehensive solution for monitoring environmental conditions and exploring historical trends. Its scalable architecture, user-friendly interface, and collaborative features make the platform suitable for researchers, students, environmental analysts, and the general public. Overall, the project demonstrates an effective application of data-driven technologies to support environmental awareness and informed decision-making.

Thank You
THANK YOU